

AI623 Final Group Project Guidelines

Deep Vision Language Models • Spring 2026

1. Project Overview

The final project for AI623 will be completed in groups of up to four students. The goal is to carry out a focused research-style study connected to the themes of the course, such as deep generative modeling, diffusion and flow-based methods, controllable generation, multimodal representation learning, unified tokenization, and vision-language modeling.

A strong project will require early topic selection, literature review, experimental planning, implementation, evaluation, and careful writing. Groups are therefore expected to start early and to keep making steady progress across the semester.

2. Group Formation and Key Deadlines

Students should form groups as early as possible. Any students who do not belong to a group by the group formation deadline will be randomly allotted to a group.

Stage	Deadline	Required Submission	What We Expect
Group formation	31 March	Group roster	Finalize a group of up to 4 members . Students without a group will be randomly allotted. Excel sheet will be shared soon where you can put the group information.
Topic selection	7 April	1-page proposal	Define the exact problem, outline the proposed approach, and include a block diagram of the intended setup.
Midterm report	8 May	Preliminary report draft	A substantial draft of the final report covering all work completed so far.
Final evaluation	1-2 June	Final report + 5-minute presentation + individual Viva	Submit the complete project. The final evaluation will be online, with an earlier in-person option if preferred.

3. Final Deliverables

Each group must complete and submit the following final deliverables:

- A final report written in the style of a conference paper. Any standard template such as NeurIPS, CVPR, ICML, or a similar venue may be used.
- A 5-minute presentation that clearly summarizes the problem, motivation, method, experiments, key results, and conclusions.

The detailed rubric for the final evaluation will be shared closer to the evaluation date.

4. Expectations for Each Stage

4.1 Topic Selection Proposal

By 7th April, each group must submit a one-page proposal. For the topic selection phase, students may refer to the buckets in the project ideas document (given below) as a rough guide to possible directions worth exploring. These buckets are intended only to suggest broad areas and are not exhaustive. Groups are also welcome to pursue their

own project idea, including one that does not fall under any of the listed buckets, provided it is relevant to the course somehow and suitable for an in-depth research project. While choosing/submitting proposal, keep in mind:

- The exact problem you intend to solve
- A short description of how you plan to approach it
- A block diagram of the intended setup or pipeline
- The dataset, benchmark, or setting you expect to use
- How you expect to evaluate success
- Why the project is feasible given your available compute and time

4.2 Midterm Report

By 8th May, each group must submit a preliminary version of the final report covering all work completed up to that point. This should be treated as an early but serious draft of the final paper rather than as a short progress note.

- Problem statement and motivation
- Relevant prior work
- Proposed method or project direction
- Experimental setup
- Results obtained so far
- Current challenges, limitations, or negative findings
- A concrete plan for the remaining work

4.3 Final Evaluation

The final evaluation will take place on 1st and 2nd June and can be held online as well. Groups may choose to do their final evaluation earlier in person if they prefer.

During the final evaluation, each group will give a five-minute presentation and submit the final report. The presentation should communicate the full story of the project clearly and efficiently.

5. What Makes a Good Project?

A good project should ask a clear, focused, answerable question. It should be narrow enough to complete well, but substantive enough to produce meaningful analysis. In general, strong project ideas usually follow a simple pattern:

1. Reproduce one baseline carefully on a small but nontrivial benchmark.
2. Make one clear extension or modification rather than changing many things at once.
3. Evaluate beyond the main metric by including at least one stress test, robustness check, or harder setting.
4. Include failure analysis showing where the method works, where it fails, and why.

6. Scope, Compute, and Feasibility

You must choose ideas based on the compute resources they actually have available. No compute resources will be provided by the course staff to any group. This means your project should be scoped realistically. If your idea depends on repeated large-scale training runs, heavy video generation, very large models, or expensive ablations, you should narrow the scope early.

- A small, clean, and well-evaluated project is better than an ambitious project that cannot be completed properly.
- Choose datasets and baselines that are realistic given your time and hardware constraints.
- Keep comparisons fair and controlled; avoid changing many components at once.
- Do not rely only on a few qualitative examples. Good evaluation matters.

7. Project Idea Buckets

Please note:

- Bucket 1-7 are directly related to course content and topics covered. (we still have to cover Bucket 5 and 6 in the remaining classes)
- Buckets 8-13 contain ideas that are more exploratory and application oriented; we have not covered these topics in the class, but fundamentals are same.
- Under each Bucket, some possible ideas are pitched but once you read it carefully along with your own research in the directions mentioned, you will be find more ideas under that bucket, on your own as well.

Bucket 1: Few-step generation and inference acceleration: Standard diffusion and many FM models still need many sampling or integration steps. Here we ask: how to make them 1-step to few-step without giving up too much quality, diversity, or controllability. Recent work attacks this through one-step training from scratch, post-hoc shortcutting of pretrained FM models, and flow-map distillation. One project could be to compare 1-step, 2-step, and 4-step generation under the same backbone and training budget. Another: to add a trajectory-consistency or self-distillation loss and test whether it reduces drift. A third could be to study when few-step models fail: rare classes, difficult prompts, or high guidance.

Expectations: Reproduce one baseline on a small image dataset. Then change one axis only: number of steps, distillation loss, trajectory consistency term, or adaptive step schedule. Report both quality and speed. The big trap is unfair comparison. Students often compare a 1-step method against a much weaker or differently trained baseline. The second trap is optimizing only FID-like quality while ignoring diversity or controllability. The third is hidden compute: some “fast” methods move the cost into distillation.

Starter references.

- [1] X. Cai *et al.*, “Shortcutting Pre-trained Flow Matching Diffusion Models is Almost Free Lunch,” NeurIPS 2025.
- [2] Z. Geng *et al.*, “Mean Flows for One-step Generative Modeling,” NeurIPS 2025.
- [3] A. Sabour, S. Fidler, and K. Kreis, “Align Your Flow: Scaling Continuous-Time Flow Map Distillation,” arXiv:2506.14603, 2025.
- [4] T. Zhang *et al.*, “FlowConsist: Make Your Flow Consistent with Real Trajectory,” arXiv:2602.06346, 2026.

Bucket 2: Reward alignment, preference optimization, and post-training control: Here the generator already exists. The task is to adapt it toward human preferences, downstream rewards, or constraints without retraining from scratch. The recent trend is to treat reward alignment as a property of the inference procedure or flow-map design itself. Good ideas could be: sparse versus dense rewards design, ODE-like versus stochastic alignment, whether reward adaptation harms diversity, if better reward-model scores actually improve human judgment on a small blind evaluation.

Expectations: Start with a pretrained generator and a simple reward, such as CLIP score, aesthetic score, or a task-specific classifier score. Compare terminal-only reward against denser stepwise reward or compare a deterministic flow-based alignment method against a stochastic one. Include qualitative failures where the reward goes up but the sample gets worse. Reward hacking is the central problem. Another challenge is that alignment results can depend heavily on the reward model itself. A third is exploration: too little stochasticity makes adaptation brittle; too much damages fidelity.

Starter references.

- [1] P. Holderrieth *et al.*, “GLASS Flows: Efficient Inference for Reward Alignment of Flow and Diffusion Models,” ICLR 2026 Oral.
- [2] H. Deng *et al.*, “DenseGRPO: From Sparse to Dense Reward for Flow Matching Model Alignment,” ICLR 2026

Poster.

[3] P. Holderrieth *et al.*, “Diamond Maps: Efficient Reward Alignment via Stochastic Flow Maps,” arXiv:2602.05993, 2026.

Bucket 3: Inverse problems, posterior sampling, and Bayesian reconstruction:

Use diffusion / FM as priors or samplers for recovering signals from corrupted measurements: deblurring, super-resolution, sparse-view reconstruction, and physics-constrained inverse problems. The recent shift is toward faster FM-style posterior samplers, unknown forward models, and gradient-free or physics-aware formulations. You may: compare diffusion-guided versus FM-guided posterior sampling, study blind or semi-blind inverse problems, build a physics-aware conditional FM baseline for a simple PDE or measurement model, compare best sample, posterior mean, and diversity metrics.

Expectations: Pick one inverse task only. Reproduce one baseline. Then test one of these: forward-model mismatch, reduced measurements, gradient-free versus gradient-based guidance, or posterior diversity versus point accuracy. Show reconstructions, uncertainty behavior, and runtime. Students often mistake good PSNR for good posterior sampling. Another issue is that inverse methods quietly assume a correct forward model. Once the forward model is wrong, performance can collapse. In scientific settings, physical validity matters as much as image quality.

Starter references.

- [1] J. Kim, B. S. Kim, and J. C. Ye, “FlowDPS: Flow-Driven Posterior Sampling for Inverse Problems,” ICCV 2025.
- [2] G. Meanti *et al.*, “Unsupervised Imaging Inverse Problems with Diffusion Distribution Matching,” ICCV 2025.
- [3] H. Dou *et al.*, “Constrained Particle Seeking: Solving Diffusion Inverse Problems with Just Forward Passes,” arXiv:2603.01837, 2026.
- [4] A. Dasgupta *et al.*, “Solving physics-constrained inverse problems with conditional flow matching,” arXiv:2603.14135, 2026.

Bucket 4: Editing, localization, and fine-grained controllability: How to edit only what should change while preserving what should stay fixed: identity, background, geometry, or scene context. Recent FM/DiT work uses layer selection, KV-cache preservation, and latent semantic control to localize edits more precisely. You may work on: background-preserving object replacement, multi-object editing with locality constraints, compare attention-feature injection versus KV-cache preservation, or study how edit strength affects source preservation.

Expectations: Use one fixed editing benchmark. Reproduce one training-free baseline. Then improve **one** of these: localization, background preservation, identity preservation, or semantic disentanglement. Include both easy and hard edits, not only cherry-picked examples. The hardest issue is the tradeoff between edit strength and preservation. Another is that real-image inversion remains brittle. A third is evaluation: you should not rely only on visual examples; use at least one preservation metric and one edit-alignment metric.

Starter references.

- [1] O. Avrahami *et al.*, “Stable Flow: Vital Layers for Training-Free Image Editing,” CVPR 2025.
- [2] T. Zhu *et al.*, “KV-Edit: Training-Free Image Editing for Precise Background Preservation,” ICCV 2025.
- [3] Y. Dalva, K. Venkatesh, and P. Yanardag, “FluxSpace: Disentangled Semantic Editing in Rectified Flow Models,” CVPR 2025.

Bucket 5: Cross-Modal Alignment, Grounded Reasoning, and Multimodal Thought:

This bucket studies the gap between seeing and thinking in VLMs/MLLMs. The core problem is that many models can encode images and align them with text reasonably well, yet still fail when they must do grounded, multi-step, visually faithful reasoning. Recent work makes this gap explicit: some papers focus on improving

the alignment interface between unimodal vision and language models; others improve the perception-to-reasoning handoff inside MLLMs; and the newest line of work trains models to reason with step-level rewards or even to perform active visual reasoning, where the model decides what additional evidence it needs before answering. Good project ideas include: probing whether better cross-modal alignment really improves downstream reasoning; adding perception-aware RL or process rewards; testing visual abstraction as a better reasoning substrate than verbose text CoT; or extending passive VLM reasoning into interactive or embodied settings.

Expectations: A good student project in this bucket should start with one base VLM/MLLM and one benchmark family, reproduce one baseline faithfully, and then change only one axis: the alignment module, the perception supervision signal, the reasoning supervision signal, the process reward, the abstraction mechanism, or the active observation policy. The most important experimental discipline here is to separate perception errors from reasoning errors. Students should explicitly analyse cases where the model looked at the wrong evidence, cases where it saw the right evidence but reasoned incorrectly, and cases where longer reasoning or interaction actually helped. The main risks are over-scoping, unclear attribution of gains, and evaluation setups that conflate better perception with better cognition.

Starter references.

- [1] L. Zhang, Q. Yang, and A. Agrawal, “Assessing and Learning Alignment of Unimodal Vision and Language Models,” in Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR), 2025, pp. 14604–14614.
- [2] B. Zou et al., “VLSA: Enhancing Vision-Language Understanding via Perception and Cognition Alignment,” OpenReview, 2026.
- [3] Z. Wang et al., “Perception-Aware Policy Optimization for Multimodal Reasoning,” in International Conference on Learning Representations (ICLR), Poster, 2026.
- [4] H. Chen, X. Lou, X. Feng, K. Huang, and X. Wang, “Unveiling Chain of Step Reasoning for Vision-Language Models with Fine-grained Rewards,” in Advances in Neural Information Processing Systems (NeurIPS), 2025.
- [5] W. Zhou, X. Xiong, Y. Peng, M. Tao, C. Zhao, H. Dong, M. Tang, and J. Wang, “PhysVLM-AVR: Active Visual Reasoning for Multimodal Large Language Models in Physical Environments,” in Advances in Neural Information Processing Systems (NeurIPS), 2025.

Bucket 6: Discrete generation, language modeling, and mixed discrete–continuous

flows: This family studies how diffusion / FM can handle tokens, categorical variables, graphs, and variable-length sequences. Recent work uses edit operations, simplex-aware flow maps, and exact guidance schemes rather than forcing everything into an awkward continuous relaxation. Project ideas could be: Variable-length generation with edit-based flows, continuous versus discrete token transport, exact guidance for discrete FM or Few-step generation on categorical data.

Expectations: Use a small discrete benchmark: toy text, short sequences, molecules, or graphs. Reproduce one baseline. Then test one of these: variable length, guidance quality, step count reduction, or exact versus approximate discrete guidance. The biggest challenges are variable length, mapping continuous states back to tokens, and the still-large gap to strong autoregressive models on many language tasks. Keep scope small and evaluation clean.

Starter references.

- [1] M. Havasi, B. Karrer, I. Gat, and R. T. Q. Chen, “Edit Flows: Variable Length Discrete Flow Matching with Sequence-Level Edit Operations,” NeurIPS 2025.
- [2] J. Jo and S. J. Hwang, “Continuous Diffusion Model for Language Modeling,” NeurIPS 2025.
- [3] D. Roos *et al.*, “Categorical Flow Maps,” arXiv:2602.12233, 2026.
- [4] Z. Wan *et al.*, “Discrete Guidance Matching: Exact Guidance for Discrete Flow Matching,” ICLR 2026 / OpenReview listing.

Bucket 7: Theory: what are diffusion and flow models actually learning? This is the theory bucket: what object is actually learned, what statistical rates are possible, and why these models generalize. Recent theory papers study higher-order FM, whether diffusion nets really learn conservative score fields, and sample complexity for rectified flows. You may work on to: test conservative versus non-conservative learned fields, compare closed-form versus stochastic FM losses, compare first-order and higher-order FM on toy mixtures, or study whether rectified-flow structure gives measurable sample-efficiency gains in practice. Another popular direction is what does CFG actually learn.

Expectations: A good theory project here should isolate one claim and test it on toy distributions or small image data. It should include both a mathematical argument and an empirical sanity check. The main danger is over-scoping. You should not attempt a grand unification here. It should make one careful claim and support it both analytically and experimentally.

Starter references.

- [1] M. Su *et al.*, “High-Order Flow Matching: Unified Framework and Sharp Statistical Rates,” NeurIPS 2025.
- [2] A. B. Vuong *et al.*, “Are We Really Learning the Score Function? Reinterpreting Diffusion Models Through Wasserstein Gradient Flow Matching,” TMLR / OpenReview listing, 2025.
- [3] Q. Bertrand *et al.*, “On the Closed-Form of Flow Matching: Generalization Does Not Arise from Target Stochasticity,” arXiv:2506.03719, 2025.
- [4] H. K. Sahoo, M. Gaur, and V. Aggarwal, “Order-Optimal Sample Complexity of Rectified Flows,” arXiv:2601.20250, 2026.

Bucket 8: Diffusion–flow bridging, hybrid transports, and probability-path design: This theme studies the relationship between diffusion, FM, and bridge-based transports (read about it). The practical question is when deterministic flow transport is preferable, when stochastic bridging helps, and how to transfer models or training objectives across paradigms. You may: compare a diffusion-bridge objective against FM on one paired generation task, transfer a diffusion prior into FM finetuning, or study whether some tasks benefit from stochastic bridges more than deterministic flows.

Expectations: Do not try to solve all theory at once. Pick one controlled setting and compare two path designs under matched compute and backbone. Show how path choice affects quality, stability, or controllability. A good project needs a very specific question: path shape, stochasticity, solver behavior, or transferability. The main risk is an unfair comparison where the two paradigms are not matched in architecture or budget.

Starter references.

- [1] J. Schusterbauer *et al.*, “Diff2Flow: Training Flow Matching Models via Diffusion Model Alignment,” CVPR 2025.
- [2] K. Zhu *et al.*, “Diffusion Bridge or Flow Matching? A Unifying Framework and Comparative Analysis,” ICLR

2026 submission / OpenReview.

[3] M. Kim, “A Unified Framework for Diffusion Bridge Problems: Flow Matching and Schrödinger Matching into One,” arXiv:2503.21756, 2025.

Bucket 9: Video generation, world models, and long-horizon coherence: This theme moves from short video clips to longer coherent rollouts and interactive visual worlds. The question is how to reduce temporal drift, maintain structure over time, and incorporate actions or richer perception. Good ideas: frame-insertion versus autoregressive rollout, long-horizon drift reduction, world-model conversion from pretrained video diffusion, conditioning with depth or other perceptual channels.

Expectations: Keep the scale small. Use short low-resolution videos or a synthetic environment. Reproduce one baseline and add one improvement for long-horizon consistency, frame insertion, or action conditioning. Use temporal metrics, not only per-frame quality. This family becomes expensive fast. The main technical issue is long-horizon drift. The main experimental issue is evaluation: per-frame scores can look good while temporal consistency is poor.

Starter references.

[1] D. Gui *et al.*, “Image as a World: Generating Interactive World from Single Image via Panoramic Video Generation,” NeurIPS 2025.

[2] S. Huang *et al.*, “Vid2World: Crafting Video Diffusion Models to Interactive World Models,” ICLR 2026 Poster.

[3] T. B. Ifriqi *et al.*, “Flowception: Temporally Expansive Flow Matching for Video Generation,” arXiv:2512.11438, 2025.

[4] Z. Liu *et al.*, “WorldWeaver: Generating Long-Horizon Video Worlds via Rich Perception,” arXiv:2508.15720, 2025 / OpenReview listing available.

Bucket 10: Robotics, planning, and generative policies: Here diffusion / FM are used as multimodal policies or planners rather than image generators. The central question is whether they can model complex action distributions while remaining trainable, safe, and fast enough for control. Some ideas: flow policy optimization in online RL, vision-to-action FM policies, low-latency policies by changing the source distribution, safety correction layers using barrier functions.

Expectations: Use a small simulation benchmark. Compare against one Gaussian or diffusion-policy baseline. Evaluate return, stability, and inference cost. If the project claims safety, include an explicit violation metric. The three-way tradeoff is expressiveness, optimization stability, and latency. A second issue is that multimodal generative policies are easy to train badly. A third is that “safer-looking trajectories” are not enough; safety must be measured explicitly.

Starter references.

[1] D. McAllister *et al.*, “Flow Matching Policy Gradients,” ICLR 2026 / OpenReview listing.

[2] Z. Li, S. Tang, and N. Azizan, “Reverse Flow Matching: A Unified Framework for Online Reinforcement Learning with Diffusion and Flow Policies,” arXiv:2601.08136, 2026.

[3] D. Gao *et al.*, “VITA: Vision-to-Action Flow Matching Policy,” ICLR 2026.

[4] J. Yang, S. Jang, and S. Han, “SafeFlowMatcher: Safe and Fast Planning using Flow Matching with Control Barrier Functions,” ICLR 2026 Poster.

Bucket 11: 3D vision, geometry, scene completion, and rendering: This theme pushes FM/diffusion into geometry-aware tasks: depth, LiDAR completion, and 3D scene generation. The point is not just pretty renderings but geometric consistency across views and structures. Ideas: few-step depth estimation with FM, LiDAR scene completion under sparsity, multimodal graph-conditioned indoor scene generation, compare 2D-latent versus 3D-latent generation for consistency.

Expectations: Pick one representation only: depth map, point cloud, or 3D scene latent. Reproduce one baseline. Then improve one part: geometry consistency, style consistency, or inference speed. Report both visual and geometric metrics. Representation choice matters a lot. Good RGB renderings do not guarantee good geometry. Another challenge is evaluation: students should include Chamfer / completion / depth metrics, not only screenshots.

Starter references.

- [1] J. Li *et al.*, “CH3Depth: Efficient and Flexible Depth Foundation Model with Flow Matching,” CVPR 2025.
- [2] A. Matteazzi and D. Tutsch, “LiFlow: Flow Matching for 3D LiDAR Scene Completion,” arXiv:2602.02232, 2026.
- [3] S. Gao *et al.*, “OneWorld: Taming Scene Generation with 3D Unified Representation Autoencoder,” arXiv:2603.16099, 2026.
- [4] Z. Yang *et al.*, “FlowScene: Style-Consistent Indoor Scene Generation with Multimodal Graph Rectified Flow,” arXiv:2603.19598, 2026.

Bucket 12: Audio, speech, and audio-visual generation: Audio is a natural place for FM because diffusion quality is attractive but many-step inference is expensive. Recent work uses FM and bridge variants for speech enhancement and for unified sound/speech generation from video. Ideas: Flow-based speech enhancement under multiple distortions, Unified audio/speech generation from video, Compare bridge path design choices, or study latency-quality tradeoffs.

Expectations: Choose one task only: enhancement, dereverberation, or video-conditioned audio. Reproduce one baseline and then test robustness to unseen noise, reverberation, or compression. Include both objective metrics and audio examples. Objective metrics like PESQ/STOI are useful but not enough. Real-world distortion mismatch is the main practical problem. In unified audio-visual generation, conditioning quality matters heavily.

Starter references.

- [1] Z. Wang *et al.*, “FlowSE: Efficient and High-Quality Speech Enhancement via Flow Matching,” Interspeech 2025.
- [2] X. Cheng *et al.*, “VSSFlow: Unifying Video-conditioned Sound and Speech Generation via Joint Learning,” ICLR 2026 / OpenReview listing.
- [3] T. Cao *et al.*, “DiT-Flow: Speech Enhancement Robust to Multiple Distortions based on Flow Matching in Latent Space and Diffusion Transformers,” arXiv:2603.21608, 2026.
- [4] D. Wang *et al.*, “Rethinking Flow and Diffusion Bridge Models for Speech Enhancement,” arXiv:2602.18355, 2026.

Bucket 13: Scientific modeling, biology, materials, wireless, and domain-specific

design: This theme covers domain-specific scientific generation and estimation, where validity is governed by chemistry, biology, or physics rather than human visual preference. Recent FM papers appear in protein design, single-cell perturbation prediction, mass-spectra-conditioned molecule generation, and wireless channel estimation. Some ideas (I am not a domain specialist here; you may need some co-supervision): protein sequence design in latent space, distributional perturbation prediction in single-cell biology, spectrum-conditioned molecule generation, plug-and-play FM estimators for channel recovery.

Expectations: Pick one domain and learn its evaluation first. Reproduce one baseline. Then add one domain-aware improvement: hard constraints, physics guidance, uncertainty calibration, or efficiency. A good project here must use domain metrics, not only generic sample-quality metrics. The main problem is that plausible-looking outputs may still be scientifically wrong. These tasks often have small data, noisy supervision, and domain-specific constraints that must be respected explicitly.

Starter references.

- [1] Z. Kong *et al.*, “ProtFlow: Fast Protein Sequence Design via Flow Matching on Compressed Protein Language Model Embeddings,” arXiv:2504.10983, 2025.
- [2] C. Yu *et al.*, “scDFM: Distributional Flow Matching for Robust Single-Cell Perturbation Prediction,” arXiv:2602.07103, 2026.
- [3] J. Nie and P. Gao, “FlowMS: Flow Matching for De Novo Structure Elucidation from Mass Spectra,” ICLR 2026 paper version visible on arXiv PDF, 2026.
- [4] Y. Zhang and Q.-U.-A. Nadeem, “Channel Estimation in MIMO Systems Using Flow Matching Models,” arXiv:2601.13827, 2026.